MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

OSCILLATOR, CRYSTAL, DIGITAL, KEY CODE QVQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic crystal oscillator module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.28 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QVQ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

* 3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40. Contact pins A21 through A40 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/236-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

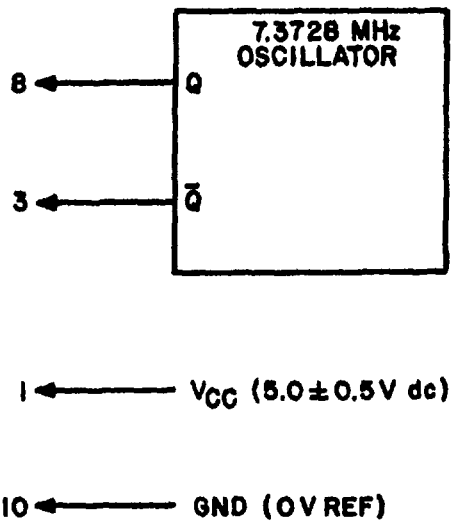
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTE: Pins 2, 4, 5, 6, 7, 9, and 11 through 40 are not used.

* FIGURE 1. QVQ module function diagram, crystal oscillator.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	V_{CC} stress	II	S	7.2728	7.4728	7.2728	7.4728	7.2728	7.4728	7.2728	7.4728	MHz
B1,B2	Isolation	III	S	10	∞	10	∞	10	∞	10	∞	MΩ
C	Power supply current	IV	X	6	60	6	60	6	60	6	60	mA
D1-D4	Frequency	V	X	7.37206	7.37354	7.37206	7.37354	7.37206	7.37354	7.37206	7.37354	MHz
E1,E2	Output voltage, high level	VI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
F1,F2	Output voltage, low level	VI	X	0	500	0	500	0	500	0	500	mV
G1,G2	Transition time (t_{TLH}), C_L = 130 pF	VI	X	2	20	2	20	2	20	2	20	ns
H1,H2	Transition time (t_{THL}), C_L = 130 pF	VI	X	2	20	2	20	2	20	2	20	ns
J	Propagation delay time (t_{PLH}), C_L = 130 pF	VI	X	2	15	2	15	2	15	2	15	ns
K	Propagation delay time (t_{PHL}), C_L = 130 pF	VI	X	2	15	2	15	2	15	2	15	ns
L1,L2	Pulse width (PW)	VI	X	61.0	74.6	61.0	74.6	61.0	74.6	61.0	74.6	ns
M	Power supply transient amplitude	VII	Q	0	700	0	700	0	700	0	700	mV
N	Power supply filter capacitor check	VIII	S	325	525	325	525	325	525	325	525	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through N shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through N in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through N shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through N shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/236 supersedes Department of the Navy, Naval Electronic Systems Command Drawing No. 28687-0102-872. Part number M28787/236-1 supersedes part numbers 0102-872 and M28787/236.

*6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	330 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Waveform	Square
Frequency	7.3728 MHz ±0.01 percent
Pulse width	67.8 ± 6.8 ns

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Output voltage	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -1 mA)	2.5 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA
High level (V_{OH} = 2.5 V)	-1 mA
Propagation delay time (Q to $\overline{Q}$ output)	
t_{PLH} (C_L = 130 pF)	15 ns maximum
t_{PHL} (C_L = 130 pF)	15 ns maximum

* 6.5 Module function. This module contains a 7.3728 MHz square wave oscillator with both a true output and a complement output.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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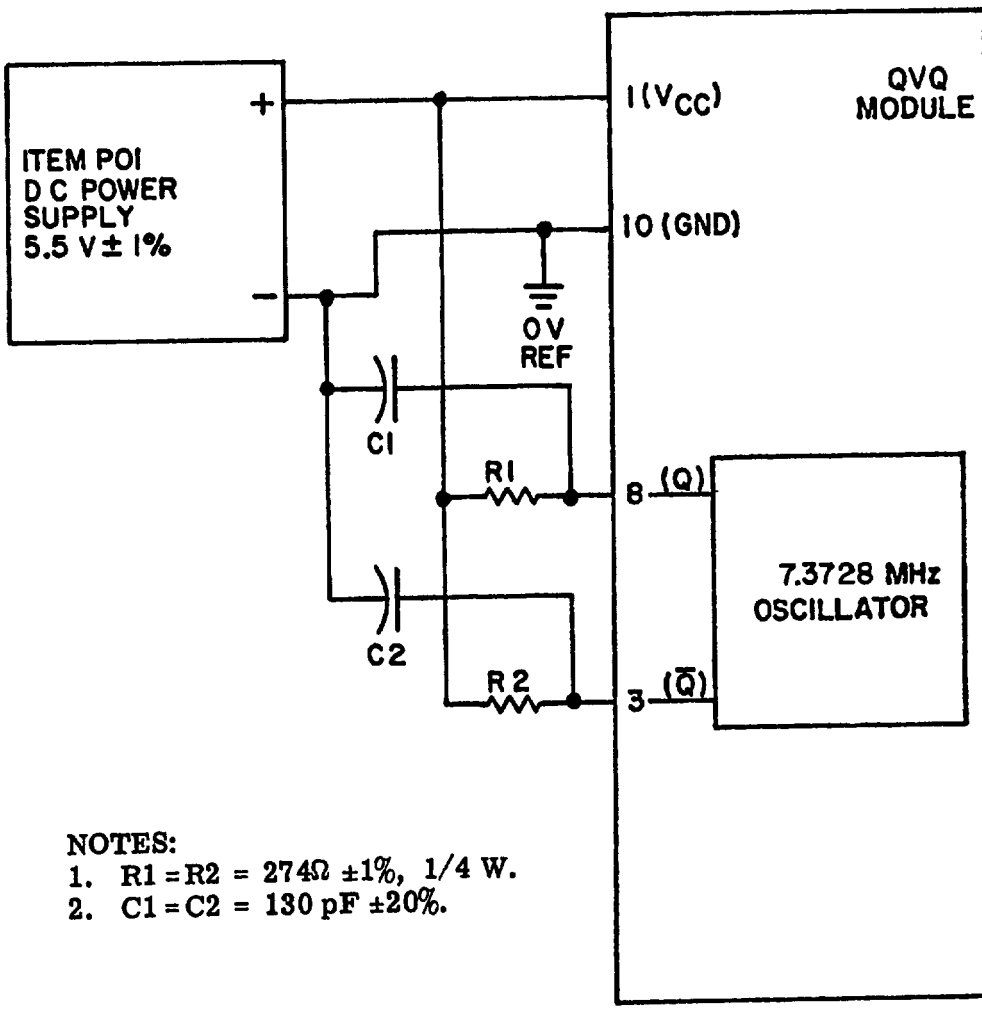


FIGURE 2. Operating temperature and life test circuit.

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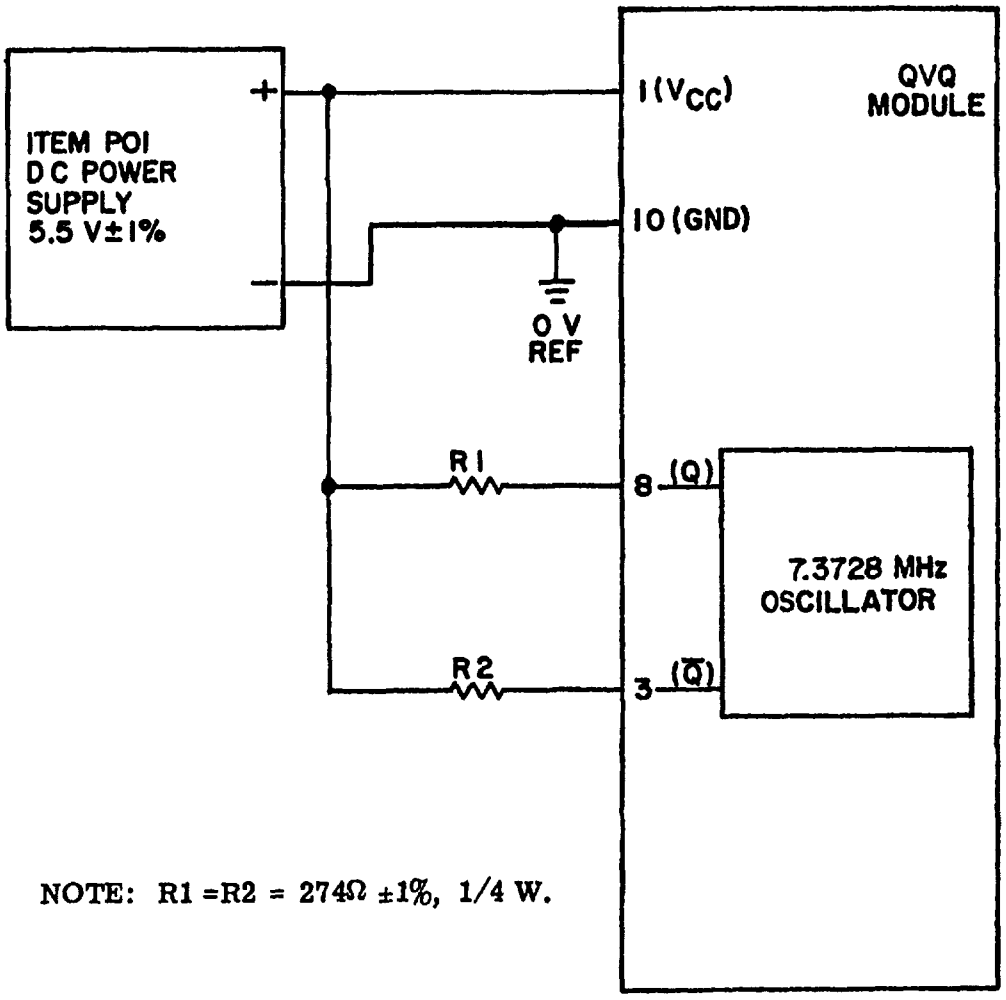


FIGURE 3. Component temperature test circuit.

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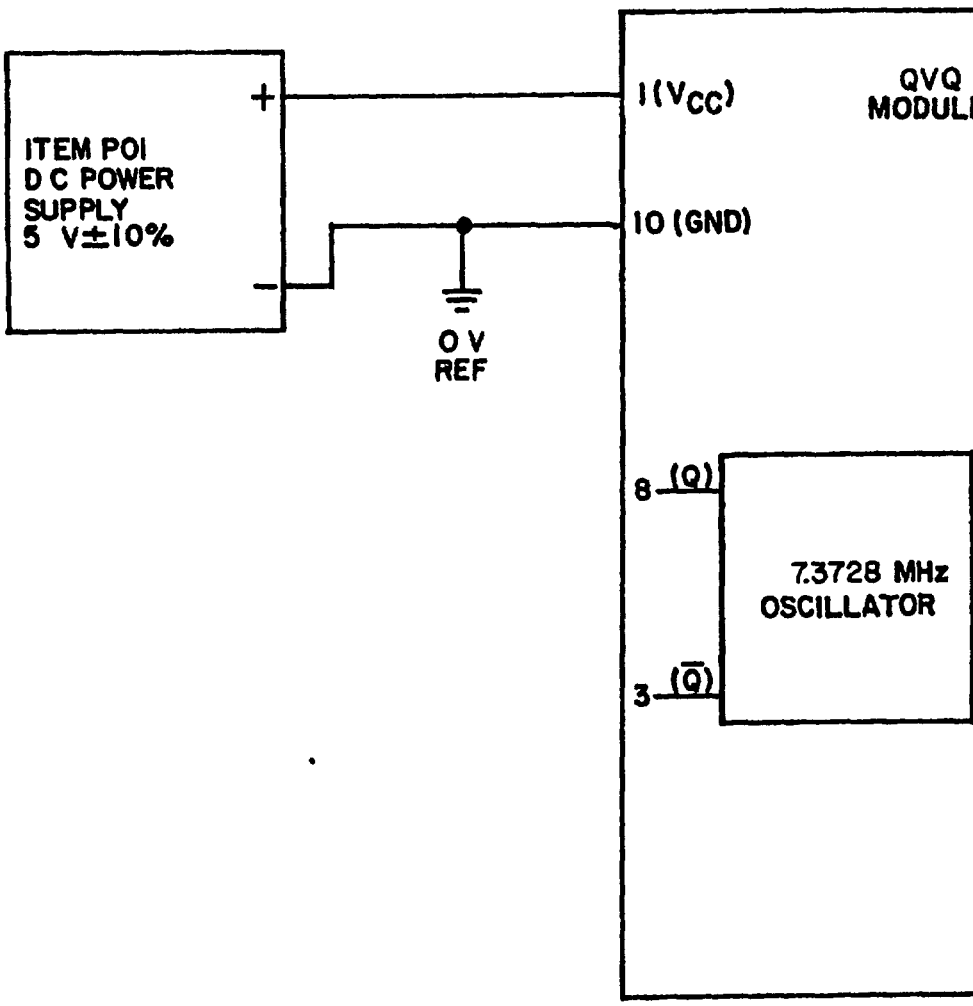


FIGURE 4. Vibration test circuit.

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TABLE II. V_{CC} stress test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	Q V_{CC} stress	8	7.2728	7.4728	MHz	5
A2	$\overline{Q}$ V_{CC} stress	3	7.2728	7.4728	MHz	5

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation, used to unused pins	1,3,8,10	All other pins	10	∞	MΩ	6
B2	Isolation, frame to module pins	All pins	Frame	10	∞	MΩ	6

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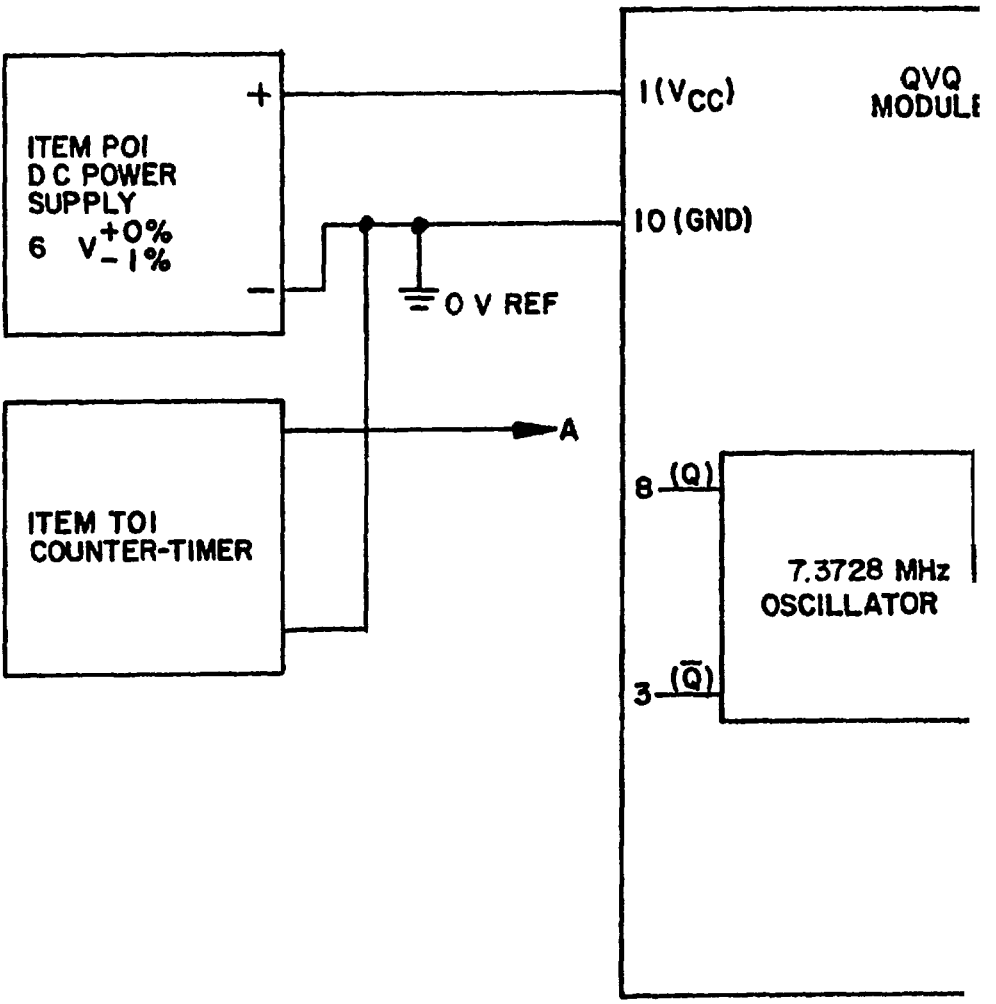
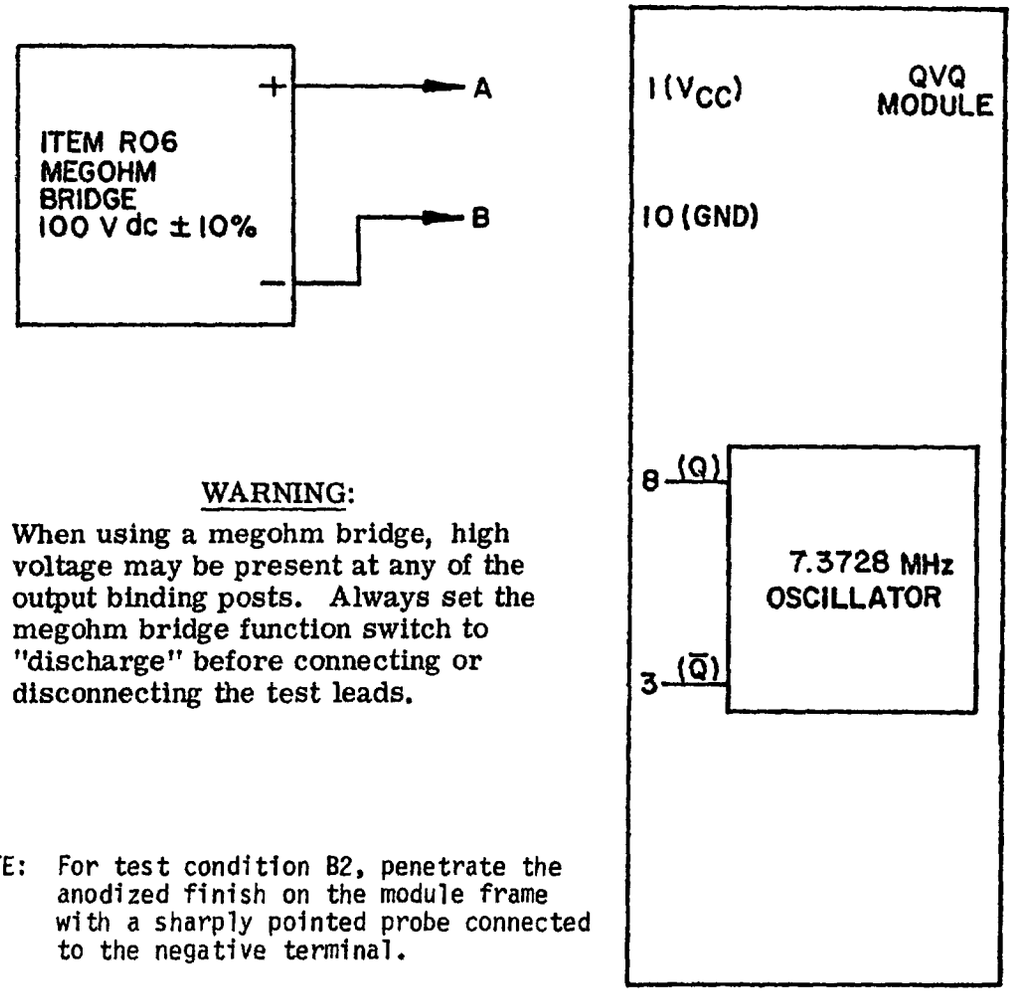


FIGURE 5. V_{CC} stress test circuit for table II.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition B2, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table III.

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TABLE IV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
C	Power supply current	6	60	mA	7

TABLE V. Frequency test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	Q Frequency	8	7.37206	7.37354	MHz	8
D2	$\overline{Q}$ "	3	"	"	"	"
D3	Q "	8	"	"	"	"
D4	$\overline{Q}$ "	3	"	"	"	"

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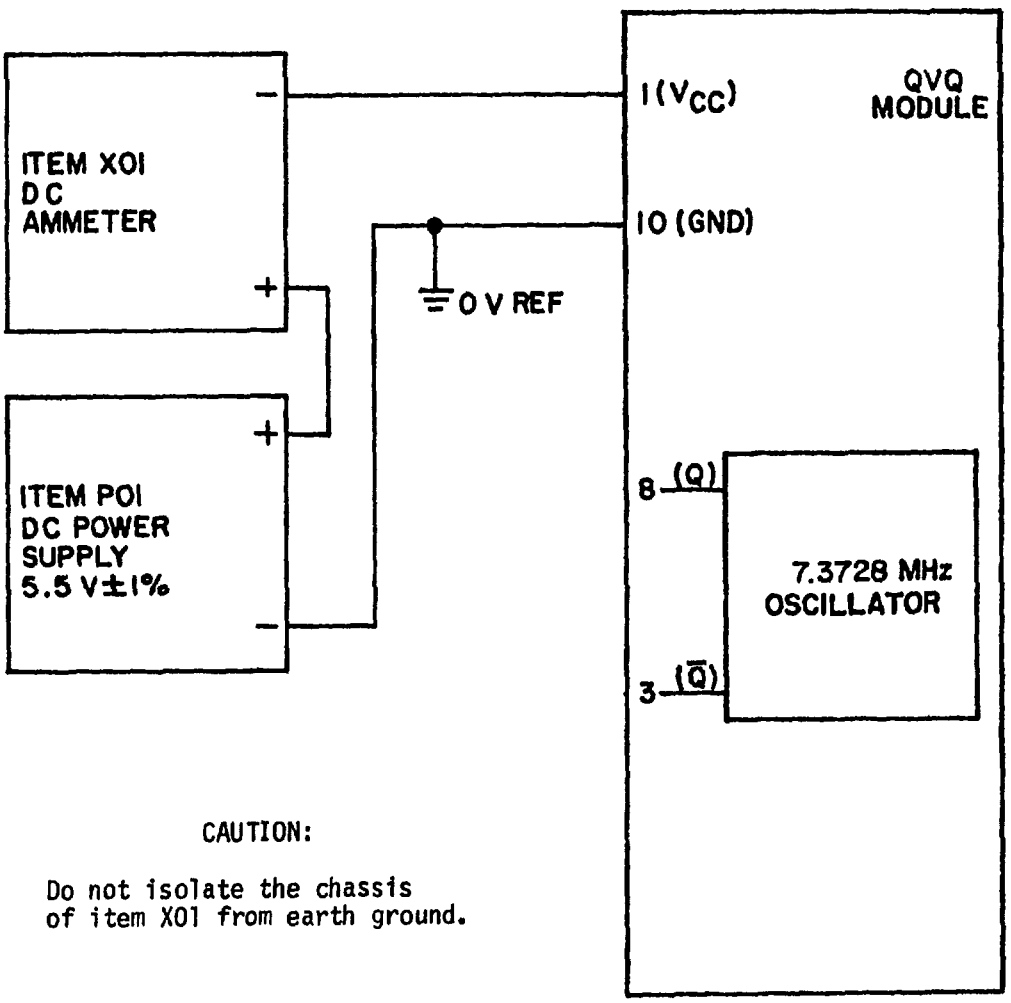
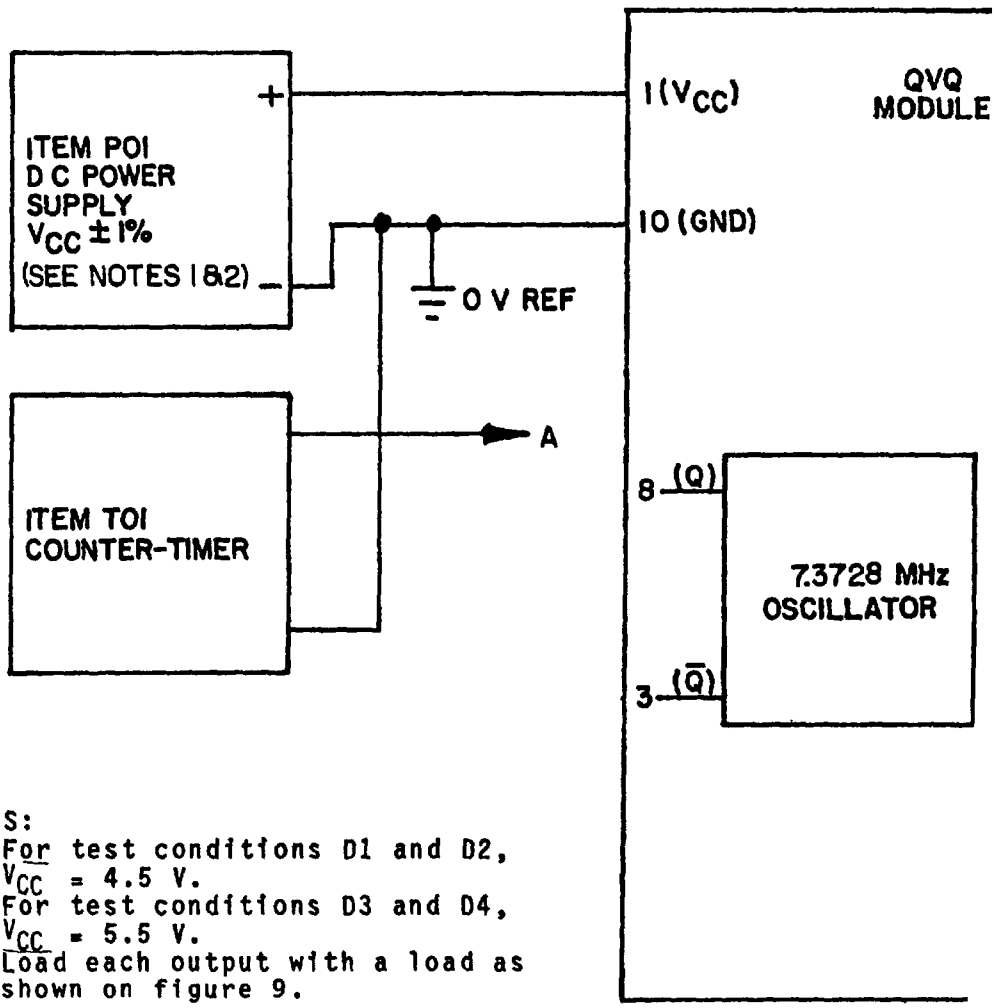


FIGURE 7. Power supply current test circuit for table IV.

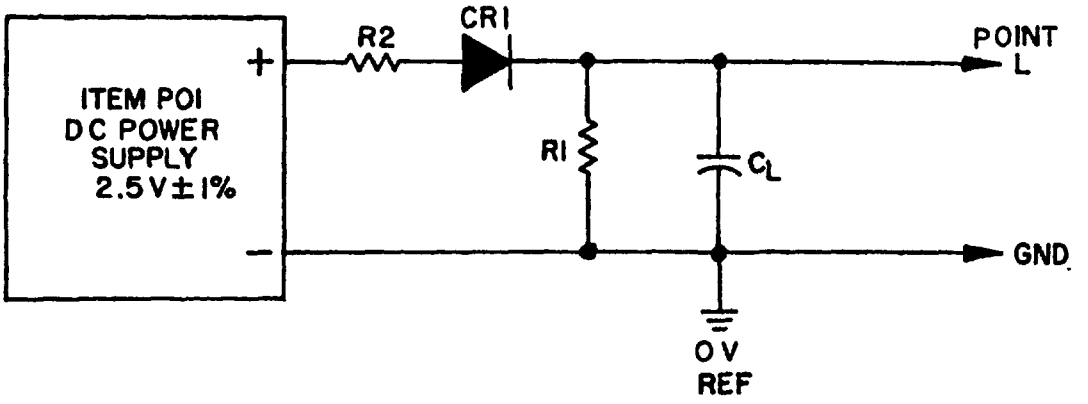
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NOTES:
1. For test conditions D1 and D2, V_{CC} = 4.5 V.
2. For test conditions D3 and D4, V_{CC} = 5.5 V.
3. Load each output with a load as shown on figure 9.

FIGURE 8. Frequency test circuit for table V.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. CL = 130 ±5 pF (including probe and parasitic capacitance).
3. R1 = 2.5 kΩ ±1%, 1/8 W.
4. R2 = Select value for a current flow of 20 mA from point L to ground.

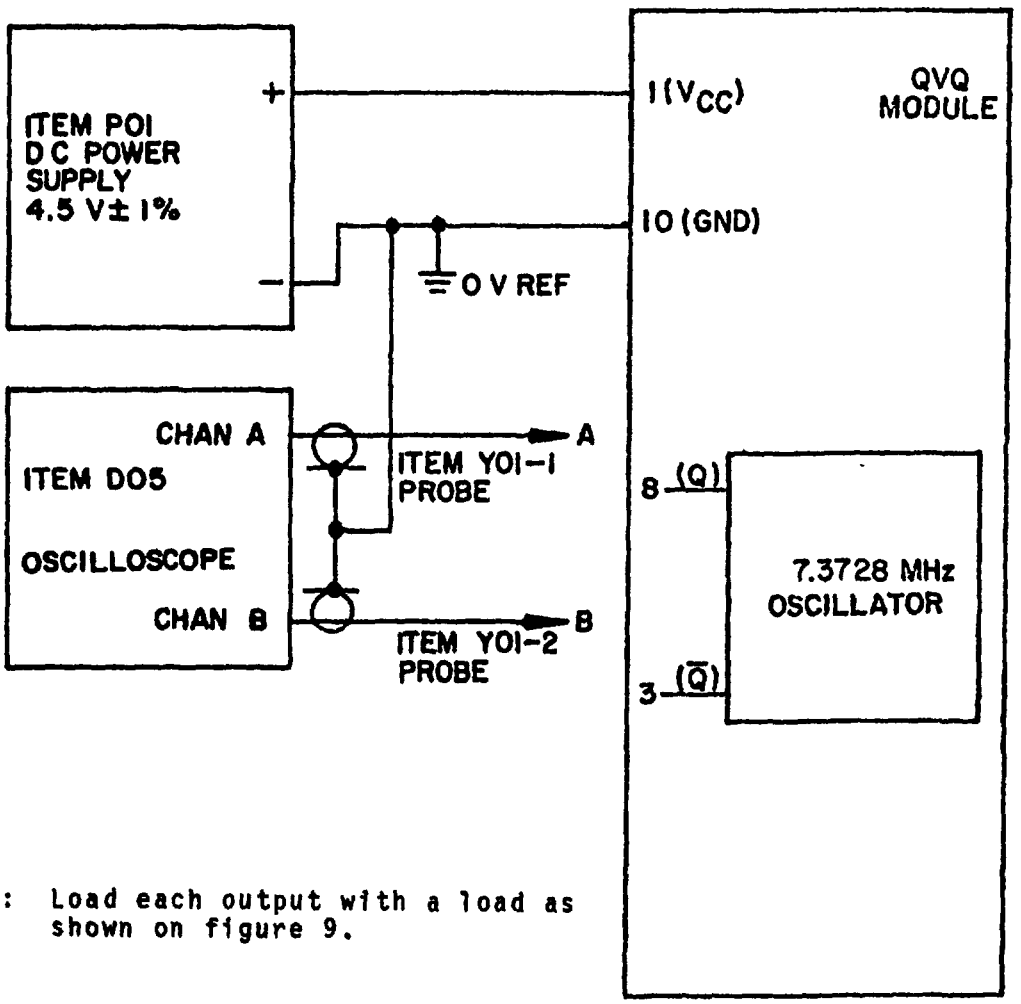
*FIGURE 9. QVQ logic load circuit.

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*TABLE VI. Output voltage (high-level and low-level), transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and pulse width test.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E1	Q	Output voltage, high level	8	Open	2.5	4.5	V	10,11
E2	$\overline{Q}$	Output voltage, high level	3	"	2.5	4.5	V	"
F1	Q	Output voltage, low level	8	"	0	500	mV	"
F2	$\overline{Q}$	Output voltage, low level	3	"	0	500	mV	"
G1	Q	Transition time (t_{TLH}), C_L = 130 pF	8	"	2	20	ns	"
G2	$\overline{Q}$	Transition time (t_{TLH}), C_L = 130 pF	3	"	"	"	"	"
H1	Q	Transition time (t_{THL}), C_L = 130 pF	8	"	"	"	"	"
H2	$\overline{Q}$	Transition time (t_{THL}), C_L = 130 pF	3	"	"	"	"	"
J	Q to $\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 130 pF	8	3	"	15	"	"
K	Q to $\overline{Q}$	Propagation delay time (t_{PHL}), C_L = 130 pF	8	3	"	15	"	"
L1	Q	Pulse width (PW)	8	Open	61.0	74.6	"	"
L2	$\overline{Q}$	Pulse width (PW)	3	Open	61.0	74.6	"	"

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NOTE: Load each output with a load as shown on figure 9.

*FIGURE 10. Output voltage (high-level and low-level), transition time (t_{TLH} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), and pulse width test circuit for table VI.

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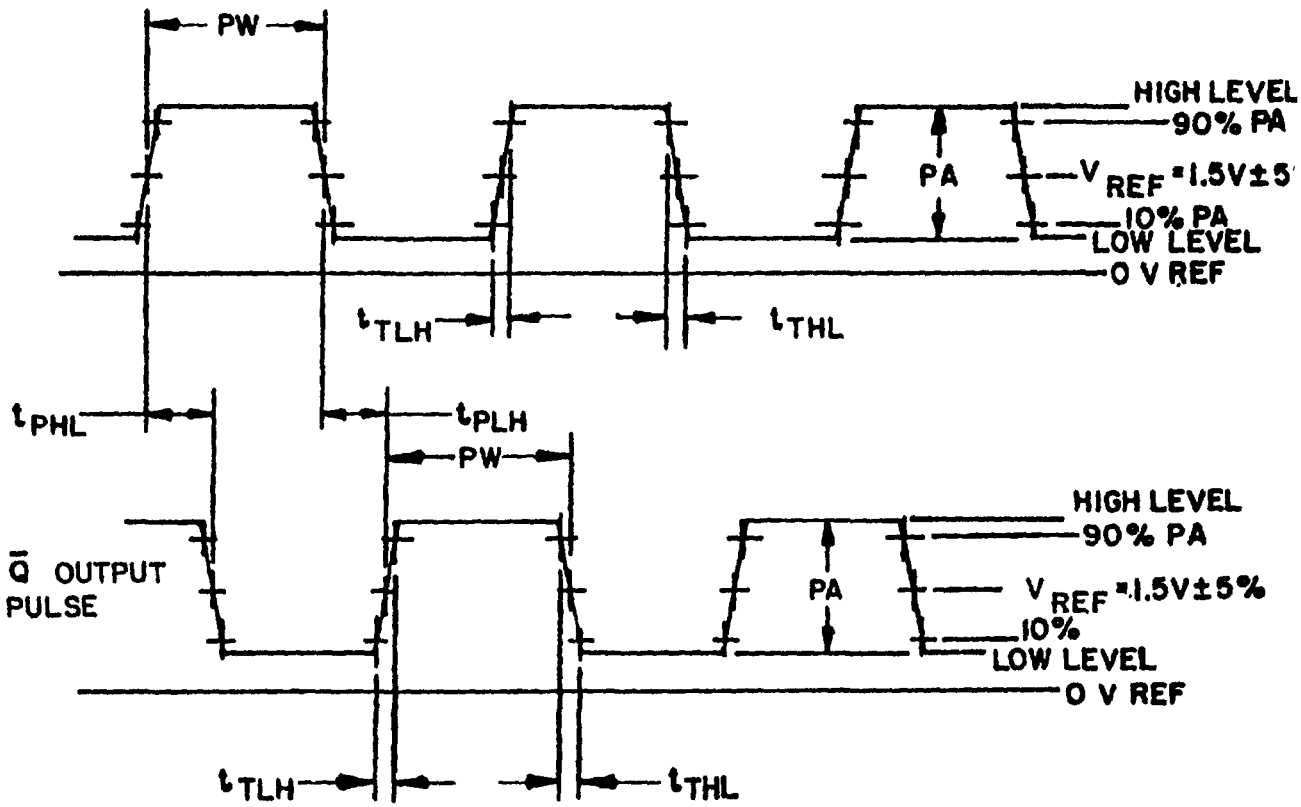


*FIGURE 11. Output voltage (high-level and low-level), transition time (t_{TLH} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), and pulse width test circuit waveforms for table VI.

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*TABLE VII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply transient amplitude	0	700	mV	12,13

TABLE VIII. Power supply filter capacitor check test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
N	Power supply filter capacitor check	325	525	mV	14,15

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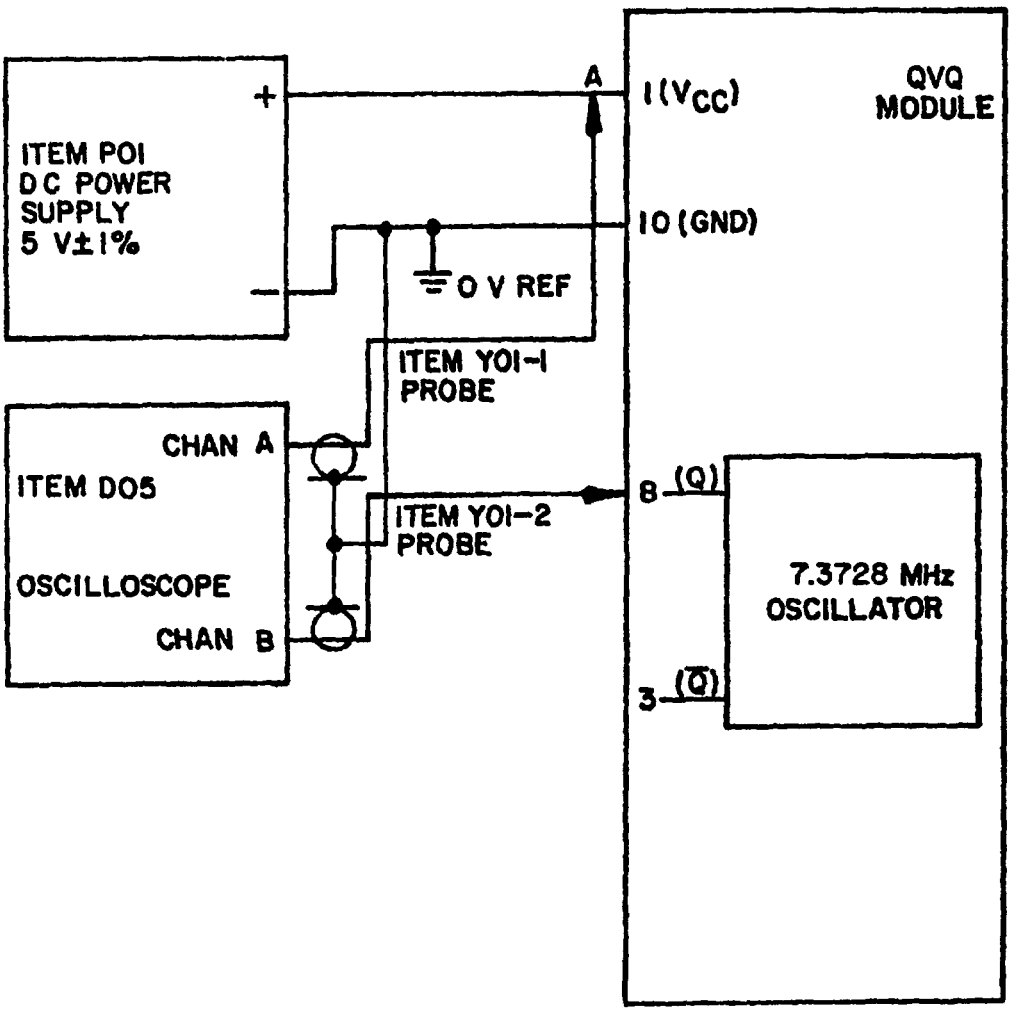


FIGURE 12. Power supply transient amplitude test circuit for table VII

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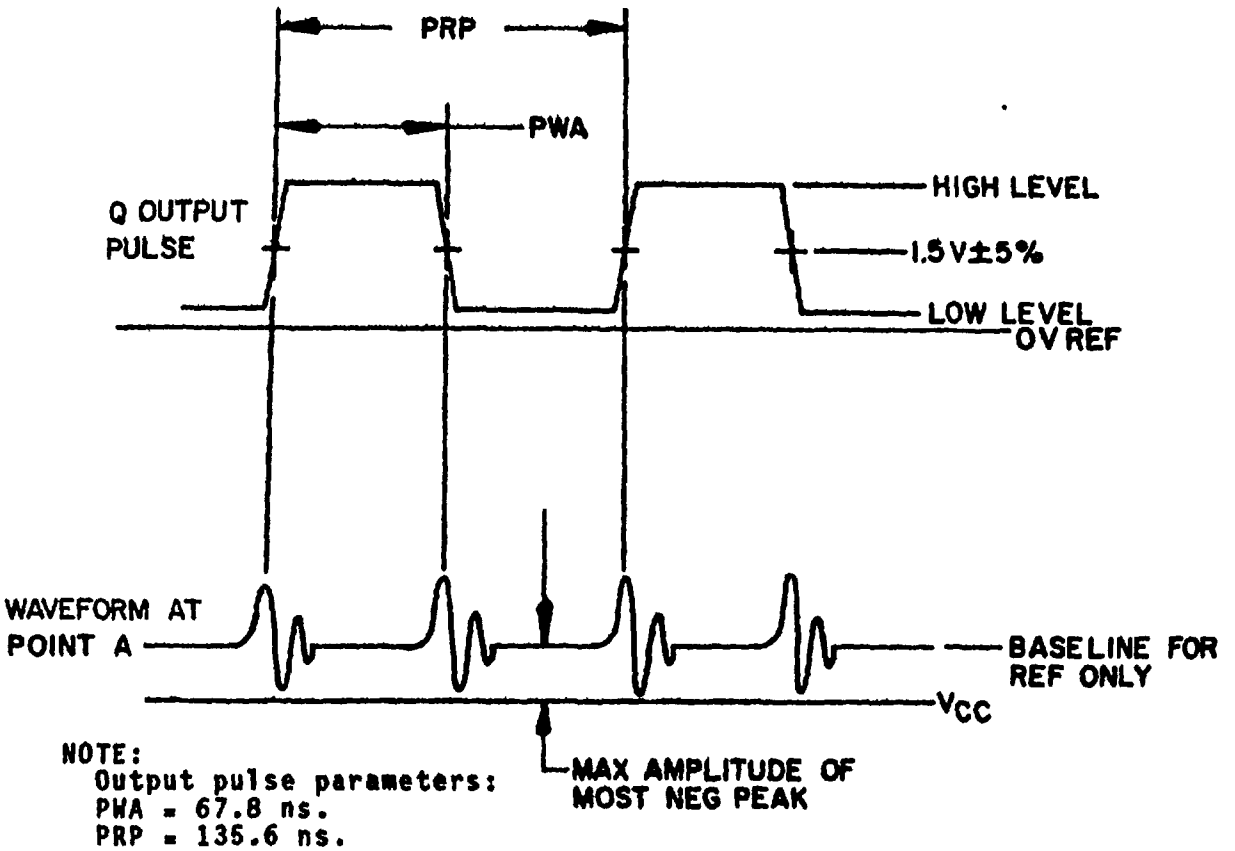
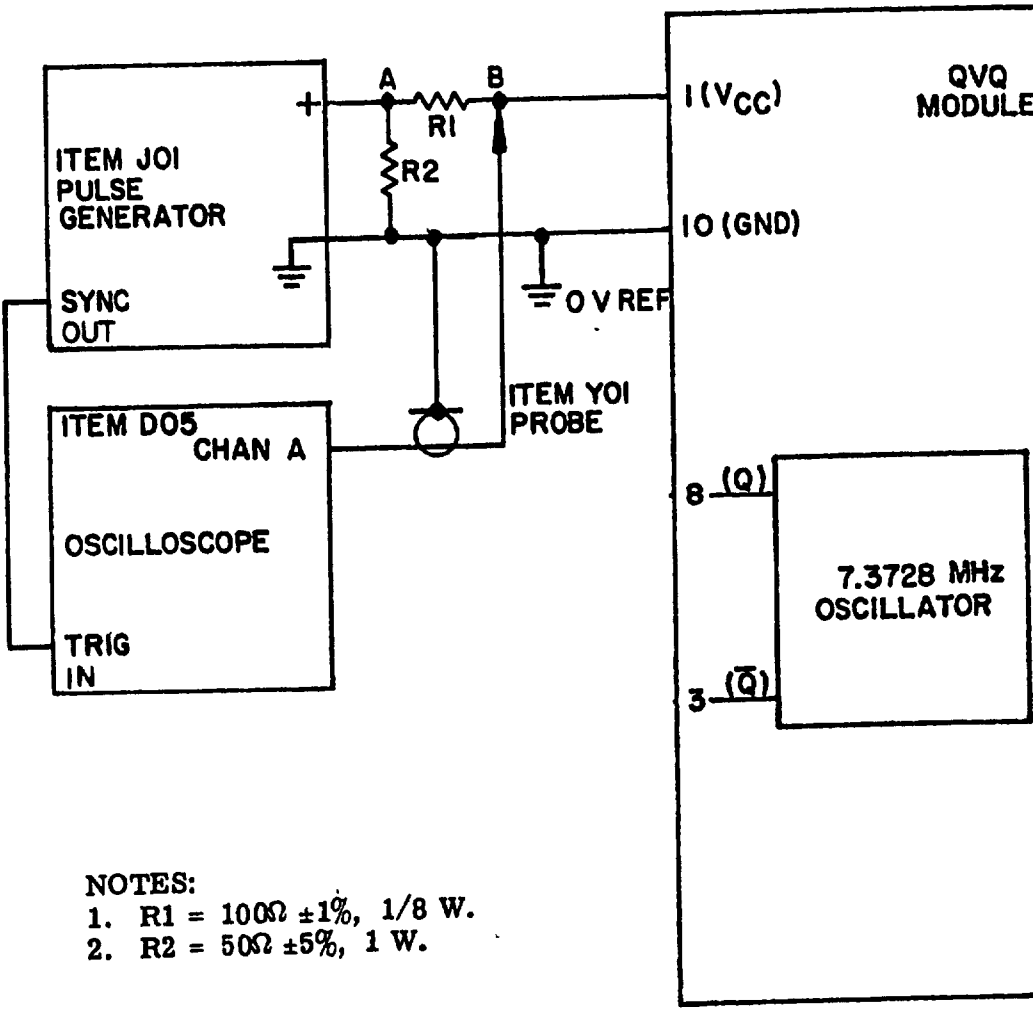


*FIGURE 13. Power supply transient amplitude test circuit waveforms for table VII.

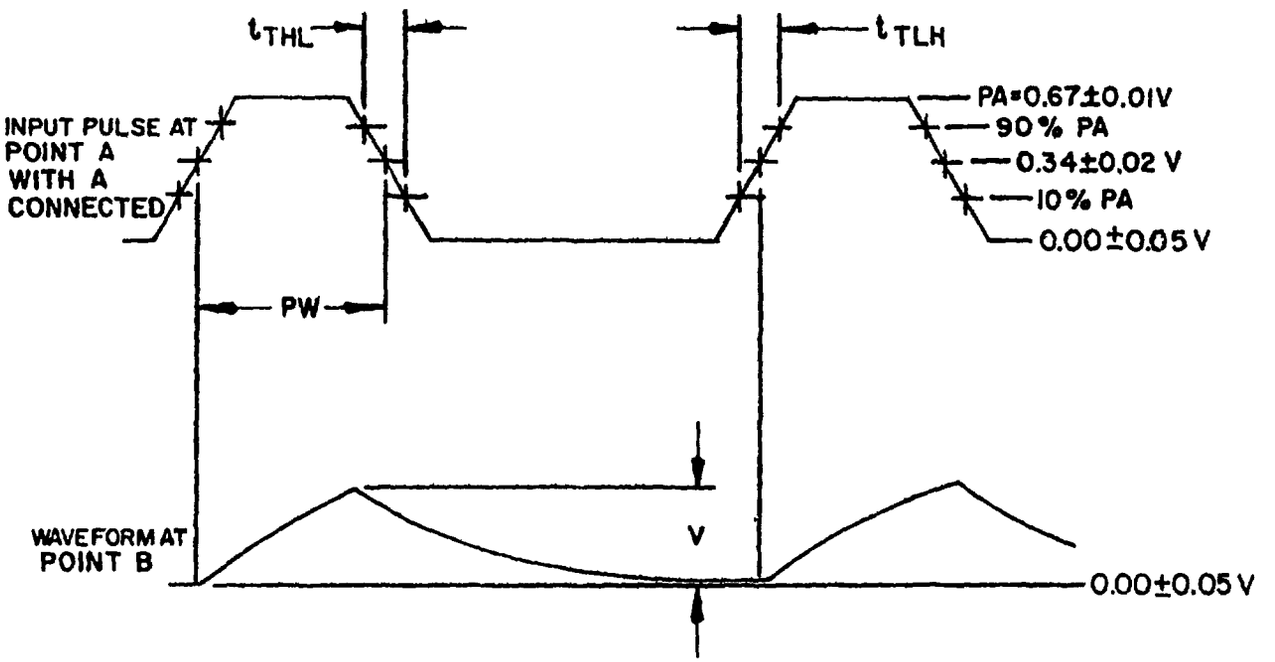
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NOTES:
1. R1 = 100Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.

FIGURE 14. Power supply filter capacitor check test circuit for table V

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NOTES:
1. Input pulse parameters:
   PW = 1.5 ±0.1 µs.
   $t_{TLH} = t_{THL}$ = 25 ±10 ns.
   PRP = 15.0 ±1.5 µs.
2. V = Peak voltage measurement.

* FIGURE 15. Power supply filter capacitor check test circuit waveforms for table VIII.

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WASHINGTON, DC 20363
ATTN: ELEX 8111

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/236A(EC)	MODULES, STANDARD ELECTRONIC OSCILLATOR, CRYSTAL, DIGITAL, KEY CODE QVQ

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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